Level 9 Project — Actuating Vice

Capstone Project

Assemblies and Mates (Level 9)

- Units: your choice (MMGS or IPS) – be consistent across all parts
- Difficulty Level: Advanced (Capstone)
- Duration: Multi-session project
- Design Intent:
  - Assembly includes a fixed jaw and a moving jaw
  - Jaw motion is driven by an actuating mechanism (e.g. lead screw, cam, or lever) – not a simple hand-pushed slide
  - Every part is fully mated; no floating or under-defined components
  - Mechanism demonstrates full range of motion (drag the mates or run a motion study)

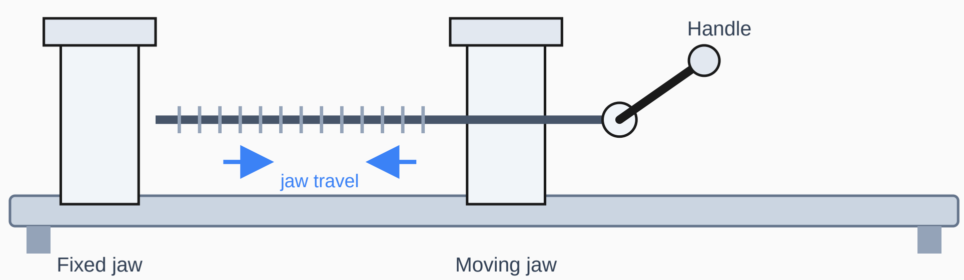
Suggested Tools / Features

- Sketch: Lines, Circles, Centerlines
- Features: Extrude Boss/Cut, Revolve, Fillets/Chamfers
- Assemblies: Concentric, Coincident, Distance, and Limit Mates
- Mechanical Mates: Screw, Cam, or Gear (pick whichever matches your mechanism)

Suggested Envelope

- Overall assembly fits within roughly 150 × 150 × 150 mm (6″ × 6″ × 6″)

HINT: Sketch your mechanism in 2D (on paper is fine) and confirm the motion works before modeling any parts in CAD.



Concept schematic only — a lead-screw mechanism is shown as one example. You choose the mechanism, dimensions, and part design that satisfy the design intent above.

Level 9 — Actuating Vice: Deliverables & Submission

What "actuating" means here

A plain vice closes because you push or turn something directly against the workpiece. An **actuating** vice closes (and ideally opens) the jaw through a built-in mechanism — a lead screw, a cam, a toggle/lever linkage, or a rack-and-pinion, for example — that a user operates without touching the jaws themselves. Choose one mechanism and design the whole assembly around it.

Requirements

- At least 4 unique machined parts (fixed jaw, moving jaw, base/frame, and at least one mechanism part), plus any fasteners
- Every part modeled as its own parametric part file, fully defined
- One assembly file with all parts fully mated (no floating components, no over-defined mates)
- The mechanism must actually move in the assembly — verify by dragging the actuator (handle, cam, lever) and confirming the jaw travels
- Reasonable, self-consistent dimensions – the parts should fit together without interference

Suggested workflow

1. Sketch the mechanism concept by hand and confirm the kinematics work
2. Model the base/frame first, then the fixed jaw
3. Model the moving jaw and the actuation parts (screw, cam, lever, etc.)
4. Assemble and mate everything, starting from the base
5. Test full range of motion before finalizing dimensions

Submission

Email your completed part and assembly files to **kainanis@hawaii.edu** — as a direct attachment, or through the **UH FileDrop** service (hawaii.edu/filedrop) addressed to the same email for larger file sets. Include your name and "Level 9 – Actuating Vice" in the subject line.

Record your submission on the course **Attendance Tracking Sheet** alongside your Levels 1–8 problems.